

WATTISHAM, United Kingdom

WMO: 035900

Lat: 52.1239N

Lon: 0.9575E

Elev: 284

StdP: 14.55

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	26.0	28.2	21.1	16.0	30.0	23.6	18.0	31.2	33.0	47.0	29.7	46.3	7.7	250	0.635

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.6	79.7	65.5	76.1	63.8	73.0	62.3	67.2	76.5	65.4	73.3	63.7	70.7	10.4	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.7	89.2	70.3	62.2	84.6	68.5	60.9	80.7	66.9	31.7	76.5	30.4	73.4	29.2	70.8	82.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 26.4	23.2	20.7	DB	20.8	86.0	3.4	3.5	18.3	88.6	16.3	90.6	14.4	92.6	11.9	95.2	(4)
(5)			WB	20.2	71.4	3.5	3.2	17.7	73.7	15.7	75.5	13.7	77.3	11.2	79.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.3	39.3	40.1	43.2	47.7	53.2	58.4	62.9	62.6	57.9	52.0	44.9	40.4		
	DBStd	10.00	5.60	5.76	5.51	5.64	5.52	4.93	4.88	4.71	4.55	5.24	5.55	6.36		
	HDD50	1476	331	280	217	108	31	1	0	0	1	38	168	301		
	HDD65	5452	795	697	675	518	368	206	102	105	217	404	604	761		
	CDD50	1579	1	2	7	39	128	253	399	392	239	100	15	4		
	CDD65	82	0	0	0	0	1	8	36	31	5	1	0	0		
	CDH74	549	0	0	0	1	6	62	254	200	23	3	0	0		
	CDH80	91	0	0	0	0	0	8	44	37	2	0	0	0		
	WSAvg	10.6	12.0	12.0	11.6	10.5	10.3	9.7	9.4	9.4	9.7	10.5	10.8	11.3		
	PrecAvg	22.9	1.9	1.5	1.7	1.7	1.7	1.9	1.9	1.9	2.1	2.2	2.4	2.1		
	PrecMax	31.0	5.2	3.7	3.7	4.3	4.1	4.5	4.0	4.4	4.7	4.9	6.1	3.9		
	PrecMin	17.3	0.2	0.2	0.3	0.2	0.2	0.2	0.2	0.1	0.0	0.1	0.7	0.6		
	PrecStd	3.6	0.9	0.7	0.9	0.9	0.8	1.1	1.0	1.1	1.3	1.3	1.1	0.9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.5	56.6	60.7	70.3	74.5	80.4	84.6	84.2	77.0	68.5	59.8	56.2		
		MCWB	51.5	51.6	51.6	56.9	61.5	65.9	66.8	66.9	63.9	60.8	56.9	53.3		
	2%	DB	52.4	53.0	57.2	64.4	70.4	75.0	80.1	79.5	72.2	64.2	57.3	54.0		
		MCWB	50.0	49.2	50.6	54.5	59.8	63.3	65.9	65.4	62.5	58.6	54.5	51.7		
	5%	DB	50.1	50.7	54.4	60.3	67.0	71.4	76.4	75.2	68.9	61.6	55.2	52.1		
		MCWB	47.8	47.4	49.2	52.1	57.6	61.6	63.9	63.6	60.5	57.4	52.7	50.0		
	10%	DB	48.0	48.6	52.0	57.3	63.4	68.2	73.1	71.7	65.9	59.7	53.1	49.9		
		MCWB	45.9	45.8	47.7	50.4	55.5	59.5	62.5	61.8	58.7	56.3	51.0	48.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.5	52.4	53.6	58.5	64.0	68.3	69.6	69.1	66.5	62.0	57.4	54.0		
		MCDB	53.9	55.4	57.7	67.3	71.9	77.8	80.7	79.8	74.3	67.2	59.0	55.4		
	2%	WB	50.2	49.9	51.7	55.3	60.9	65.1	67.3	66.6	63.9	59.9	55.1	52.3		
		MCDB	52.0	52.3	55.6	62.8	68.4	72.6	77.4	75.1	69.8	63.0	56.7	53.9		
	5%	WB	48.1	47.9	50.1	53.1	58.5	62.7	65.4	64.9	62.0	58.4	53.1	50.2		
		MCDB	49.8	50.2	53.7	59.0	65.6	69.0	73.8	72.5	66.8	60.8	54.8	51.8		
	10%	WB	46.1	46.3	48.2	51.3	56.4	60.7	63.6	63.3	60.3	56.9	51.2	48.2		
		MCDB	47.8	48.3	51.6	56.4	62.4	67.0	70.9	69.7	64.4	59.2	52.9	49.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.8	10.1	12.4	14.9	15.6	15.8	16.6	16.0	14.4	11.9	9.7	8.4		
		MCDBR	11.1	11.9	15.4	18.8	20.5	20.1	21.6	20.2	17.5	12.9	10.4	9.5		
		MCWBR	10.5	9.1	10.2	10.7	11.6	10.8	9.6	8.6	9.1	8.6	8.9	8.9		
	5% WB	MCDBR	11.0	11.1	14.3	17.6	19.2	18.6	19.5	18.1	15.7	11.7	10.0	9.6		
		MCWBR	10.7	9.4	10.4	10.7	11.5	11.2	9.9	8.6	9.2	8.9	9.1	9.2		
(40) Clear-Sky Solar Irradiance	taub		0.325	0.342	0.376	0.398	0.404	0.407	0.406	0.396	0.374	0.357	0.330	0.315		
	taud		2.369	2.381	2.278	2.237	2.265	2.277	2.293	2.342	2.398	2.429	2.413	2.355		
	Ebn at Noon		213	245	256	265	269	269	267	263	255	236	209	194		
	Edh at Noon		19	25	34	39	40	40	39	35	30	24	18	17		
(44) All-Sky Solar Radiation	RadAvg		237	451	804	1275	1535	1656	1579	1325	994	575	292	187		
	RadStd		22	67	93	128	114	129	156	97	81	53	32	23		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (79 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page